

Aerial Mining Operations - Multi-Class Object Tracking

Task type: Video Object Tracking (multi-class, frame-by-frame)

Tool: Label Studio

Source footage: 18-second aerial drone clip of an active mining/quarry site

Status: Complete

Overview

This project demonstrates multi-class object tracking on real-world aerial footage of a mining/quarry operation. The goal was to track vehicles and equipment across frames with consistent identity (track IDs), following a documented labeling standard rather than ad-hoc judgment calls.

This is representative of real-world use cases such as fleet utilization monitoring, site safety analysis, and automated equipment tracking for mining and construction operations.

Classes Tracked

Class	Count	Notes
Wheel Loader	1	Actively working the stockpile throughout the clip
Dump Truck	5	Includes both stationary and in-motion units
Pickup Truck	2	Light vehicles present on-site

All 8 objects were tracked with consistent IDs from first appearance to last visible frame.

Methodology

Full labeling rules are documented in [labeling_guidelines.md](#), covering:

- Class definitions and disambiguation rules (e.g., distinguishing dump trucks from other equipment at aerial angles)
- A 40%-visibility occlusion threshold, with special handling for dust/smoke interruptions common in mining footage
- Re-identification rules for objects exiting and re-entering frame
- Known footage challenges (dust plumes, aerial perspective, similar vehicle silhouettes from above)

The annotation process followed a two-pass approach: a rough first pass across all frames to establish every track, followed by a refinement pass to tighten bounding boxes and correct any class or drift errors before final submission.

Notable Challenges Handled

- **Aerial perspective classification:** at a top-down angle, dump trucks in motion can visually resemble flatbed equipment carriers due to foreshortening of the truck bed. This was corrected through careful review against known site equipment rather than assumed shape alone.
- **Dust plume occlusion:** a central dust/smoke plume in the footage periodically reduced visibility of nearby equipment; track continuity was maintained using the documented occlusion rule rather than dropping or duplicating IDs.
- **Small-object tracking:** a distant pickup truck near the top of frame remained accurately tracked despite its small size relative to the frame.

Files in This Submission

- [labeling_guidelines.md](#) — full labeling standard used for this project
- [annotations_export.json](#) — raw exported tracking annotations from Label Studio
- [screenshots/](#) — sample frames showing tracking in progress and final state